1(a)	$\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right) 6\times 3(3x-2)^5 \quad (=18(3x-2)^5)$	M1A1
		(2)
(b)	$"18"(3\times\tfrac{1}{3}-2)^{"5"}=-18$ $-18\rightarrow\frac{1}{18}$ $y-1="\frac{1}{18}"\left(x-\frac{1}{3}\right)$ $3x-54y+53=0$	M1 M1 dM1 A1
		(4)
		(6 marks)

2(a)	$\text{fg}(5) = \text{f}(3) = \dfrac{5-"3"}{3\times"3"+2}, = \dfrac{2}{11}$	M1, A1
		(2)
(b)	$\text{f}^{-1}(x) = \dfrac{5-2x}{3x+1}$	M1A1
	$x \in \mathbb{R},\ x \neq -\dfrac{1}{3}$	B1
		(3)
(c)	$\dfrac{5-\frac{1}{a}}{3\times\frac{1}{a}+2} = 2(a+3)-7$	B1ft
	$\dfrac{5-\frac{1}{a}}{3\times\frac{1}{a}+2} = 2(a+3)-7 \Rightarrow 4a^2-a-2=0$	M1A1
	$(a=)\ \dfrac{1\pm\sqrt{33}}{8}$	A1
		(4)
		(9 marks)

3(a)	$\int \frac{9x}{3x^2+k}\,\mathrm{d}x = \frac{3}{2}\ln\left(3x^2+k\right)+C$	M1A1
		(2)
(b)	$\frac{3}{2}\ln(75+k)-\frac{3}{2}\ln(12+k)=\ln 8$	M1
	$\frac{3}{2}\ln\left(\frac{75+k}{12+k}\right)=\ln 8$ oe	dM1
	$\frac{75+k}{12+k}=4 \Rightarrow k=...$	ddM1
	$(k=)\ 9$	A1
		(4)
		(6 marks)

4(a)	Gradient $= \dfrac{3.85-3.08}{5-0}$ (=0.154 or $\dfrac{77}{500}$) $\log_{10} N = 3.08 + 0.154t$ oe	M1 A1
		(2)
(b)	$a = 10^{3.08} =$ awrt 1200 $\log_{10} b =$ "0.154" or $b = 10^{\text{"0.154"}}$ $b =$ awrt 1.43	B1 M1 A1
		(3)
(c)	$\log_{10} 500\,000 = 3.08 + 0.154T \Rightarrow T = \ldots$ or $T = \log_{\text{"1.43"}}\left(\dfrac{500\,000}{\text{"1200"}}\right) = \ldots$ $(T =)$ awrt 17	M1 A1
		(2)
		(7 marks)

5(a) **(i)**	7	B1
(ii)	$\frac{9}{k}$	B1
		(2)
(b)	$kx-9-2=0$ and $-kx+9-2=0 \Rightarrow \frac{\dots}{k}, \frac{\dots}{k}$	M1
	CVs $\frac{7}{k}, \frac{11}{k}$	A1
	$\frac{7}{k} < x < \frac{11}{k}$	A1
		(3)
(c)	$"-2" = -2 \times "\frac{9}{k}" + 3 \Rightarrow k = \dots$	M1
	$(k =)\ 3.6$	A1
	$k > 3.6$	A1
		(3)
		(8 marks)
Alt(b)	$-2 < kx - 9 < 2$	
	$\frac{\dots}{k} < x < \frac{\dots}{k}$	M1
	$\frac{7}{k} < x < \frac{11}{k}$	A1A1

6(a)	$\left(\mathrm{f}'(x)=\right) 5(x^2-2)\times\frac{1}{2}\times 4\times(4x+9)^{-\frac{1}{2}}+10x(4x+9)^{\frac{1}{2}}$ $\left(\mathrm{f}'(x)=\right) \frac{10(x^2-2)+10x(4x+9)}{(4x+9)^{\frac{1}{2}}}=\frac{50x^2+90x-20}{(4x+9)^{\frac{1}{2}}}$ $\left(\mathrm{f}'(x)=\right) \frac{10(5x^2+9x-2)}{(4x+9)^{\frac{1}{2}}}$	M1A1 dM1 A1
		(4)
(b)	Sets $\mathrm{f}'(x)=0$ to give $x=-2, \frac{1}{5}$	B1
		(1)
(c)	$y=5\left(("-2")^2-2\right)\left(4\times("-2")+9\right)^{\frac{1}{2}}=10$ $(-2, 10)$	M1 A1cao
		(2)
(d)	$\mathrm{f}(0)=-30$ $\text{Upper bound}=2\times"10"+4=\ldots$ or $\text{Lower bound}=2\times"-30"+4=\ldots$ $-56\leqslant \mathrm{g}(x)\leqslant 24$	B1 M1 A1
		(3)
		(10 marks)

7(a)	$3\cot^2 2\theta - 7 = \dfrac{13}{2\sin\theta\cos\theta}$ **or** $2\sin\theta\left(3(\operatorname{cosec}^2 2\theta - 1) - 7\right) = 13\sec\theta$ **oe**	M1
	$3(\operatorname{cosec}^2 2\theta - 1) - 7 = \dfrac{13}{2\sin\theta\cos\theta}$	dM1
	$\Rightarrow 3\operatorname{cosec}^2 2\theta - 10 = 13\operatorname{cosec} 2\theta \Rightarrow 3\operatorname{cosec}^2 2\theta - 13\operatorname{cosec} 2\theta - 10 = 0$*	ddM1A1*
		(4)
(b)	$3\operatorname{cosec}^2 2\theta - 13\operatorname{cosec} 2\theta - 10 = 0$	
	$(3\operatorname{cosec} 2\theta + 2)(\operatorname{cosec} 2\theta - 5) = 0 \Rightarrow (\operatorname{cosec} 2\theta) = 5 \Rightarrow (\sin 2\theta) = \dfrac{1}{5}$	M1A1
	$\theta = \dfrac{\sin^{-1}\left("\pm\frac{1}{5}"\right)}{2} = \ldots$	dM1
	$(\theta =)$ awrt 0.101, 1.47	A1
		(4)
		(8 marks)
Alt(a)	$2\sin\theta\left(\dfrac{3(1-\sin^2 2\theta)}{\sin^2 2\theta} - 7\right) = 13\sec\theta$	M1
	$\dfrac{3(1-\sin^2 2\theta)}{\sin^2 2\theta} - 7 = \dfrac{13}{2\sin\theta\cos\theta}$	dM1
	$\dfrac{3}{\sin^2 2\theta} - 10 = \dfrac{13}{\sin 2\theta}$	ddM1
	$3\operatorname{cosec}^2 2\theta - 13\operatorname{cosec} 2\theta - 10 = 0$ *	A1*

8(a)	$\dfrac{\mathrm{d}v}{\mathrm{d}t} = -\mathrm{e}^{t-10} + 9\mathrm{e}^{-0.75t} = 0 \Rightarrow 9 = \mathrm{e}^{1.75t-10}$ or $9\mathrm{e}^{-0.75t} = \mathrm{e}^{t-10}$	M1A1
	$9 = \mathrm{e}^{1.75t-10} \Rightarrow t = \dfrac{10+\ln 9}{1.75}$ or $\ln 9 - 0.75t = t - 10 \Rightarrow t = \dfrac{10+\ln 9}{1.75}$	M1
		A1
	$t =$ awrt 6.97	A1
	awrt 11.9 (ms^{-1})	
		(5)
(b)	$\int 12 - 12\mathrm{e}^{-0.75t} - \mathrm{e}^{(t-10)}\,\mathrm{d}t = 12t - \mathrm{e}^{(t-10)} + 16\mathrm{e}^{-0.75t}$ $(+C)$	M1A1
	$\left[12t - \mathrm{e}^{(t-10)} + 16\mathrm{e}^{-0.75t}\right]_0^T = 100 \Rightarrow 12T = \ldots$	M1
	$T = \dfrac{1}{12}\left(116 - 16\mathrm{e}^{-0.75T} + \mathrm{e}^{T-10} - \mathrm{e}^{-10}\right)$ *	A1*
		(4)
(c) **(i)**	$T_2 = \dfrac{1}{12}\left(116 - 16\mathrm{e}^{-0.75\times 10} + \mathrm{e}^{10-10} - \mathrm{e}^{-10}\right)$	M1
	$T_2 =$ awrt 9.7493	A1
(ii)	9.7293 (seconds)	A1
		(3)
		(12 marks)

9(a)	$\frac{dy}{dx} = \frac{(1+\sin x)(-2\sin x) - (1+2\cos x)\cos x}{(1+\sin x)^2}$	M1A1
	$(1+\sin x)(-2\sin x) - (1+2\cos x)\cos x = 0$ $\Rightarrow -2\sin x - \cos x - 2 = 0$	dM1
	$2\sin x + \cos x = -2$ *	A1*
		(4)
	Possible solutions from $R\sin(x+\alpha)$ or $R\cos(x-\alpha)$: $2\sin x + \cos x = -2 \Rightarrow \sqrt{5}\sin(x+0.464) = -2$ or $\sqrt{5}\cos(x-1.107) = -2$ or eg solutions from squaring	M1M1A1
	$(2\sin x + \cos x)^2 = 4 \Rightarrow \cos x(3\cos x - 4\sin x) = 0$ --- $\sqrt{5}\sin(x+0.464) = -2$ or $\sqrt{5}\cos(x-1.107) = -2 \Rightarrow x = ...$ or $\tan x = \frac{3}{4} \Rightarrow x = ...$	dM1
	$x =$ awrt 3.79	A1
		(5)
		(9 marks)